Harivijay R Gunaseelan

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science (BASc), Computer Engineering | 2025 - Present

- **Coursework:** Digital Computers, Numerical Methods, C++ (DSA, OOP), Embedded Systems (Arduino, sensors)

Skills

Languages: Python, C++, Embedded C++ (Arduino), MATLAB, JavaScript, C#, HTML, CSS
AI/ML Tools: LangChain, FAISS, HuggingFace Embeddings, Scikit-learn, Pandas, RAG Pipelines, Groq API
Backend & APIs: FastAPI, Django, REST APIs, Webhooks, CORS, Cloudflare Tunnel
Hardware & Embedded: ESP32, I2C , Arduino Nano, Biomedical Sensors (SpO_2, Accelerometer)
Tools: Git, GitHub, VS Code, Render, Vercel, React, Arduino IDE, Unity, Bootstrap, CI/CD pipelines

Work Experience

Accessibility Review Team Member — **University of Waterloo**
Waterloo, ON (Remote, Project-Based) June 2026 - August 2026

- Audited **HTML source code** across **3 UW PD courses** against **WCAG accessibility standards**, identifying and documenting content quality issues impacting assistive technology users across multiple course modules.
- Produced **structured compliance reports** with cited WCAG criteria for each finding, delivering clear, verifiable, and actionable documentation that enabled stakeholders to resolve accessibility gaps efficiently.
- Aligned classification and validation standards through recurring team syncs and created structured, screen-reader-friendly transcripts translating visual course content into accessible text.

Projects

AI GitHub PR (Pull Request) Review Bot
View Repository | *Personal Project* | View in Portfolio April 2026 - May 2026

- Built an **AI-driven PR review bot** using **FastAPI** and **LangChain**, integrating **GitHub Webhooks** to automatically trigger code reviews on every new PR without any manual intervention.
- Fetched PR diffs via **GitHub REST API** and constructed **structured LangChain prompt chains** to route code changes through **Groq (LLaMA 3.3)**, generating categorized feedback on bugs, readability, and improvements.
- Exposed the **FastAPI backend** to GitHub's webhook delivery system using a **Cloudflare Tunnel**, enabling fully automated end-to-end PR reviews to complete within **10–15 seconds** of a pull request being opened.

ENGHacks Hackathon — Goose Hunt Multiplayer Web Game
View Repository | *Front-End Developer* | View in Portfolio March 2026 (12 hours)

- Delivered a **JavaScript** front-end for a real-time location-based multiplayer game within a **12-hour hackathon** as part of a **4-person team**, shipping a fully functional and playable product under a hard deadline.
- Integrated **Leaflet.js API** for live geolocation tracking and implemented a **dynamic shrinking play-zone mechanic** that updates in real-time as players move, creating a battle royale-style core gameplay loop.
- Built an **interactive multiplayer lobby system** with session creation and **shareable join codes**, implementing navigation across **4+ pages** and connecting frontend gameplay UI with REST backend services built by teammates.

DocuMind — AI Document Q&A Platform
View Repository | *Personal Project* | Project Link | View in Portfolio April 2026 - May 2026

- Built a **full-stack RAG application** (**React, Tailwind CSS, FastAPI**) supporting **PDF, DOCX, and XLSX** uploads, enabling natural-language queries over multi-page documents without manual searching.
- Designed an end-to-end retrieval pipeline (ingestion → chunking → **HuggingFace** embeddings → **FAISS** vector search → **Groq/LangChain** generation) returning context-aware answers grounded in uploaded document content.

Achievements & Certifications

- **FreeCodeCamp Python Developer Certification**: 300-hour certification in Python fundamentals, OOP & DSA.
- **MATLAB Onramp** (MathWorks): Certification in MATLAB syntax, matrix operations, data import and plotting